

Recycled Concrete Base vs. Virgin Aggregate

Project lead and lead analyst | University of Windsor | 2026

A screening-level cradle-to-site comparison assessed material production and transport for a 20-km, 7.5-m-wide, 0.25-m-thick road subbase.

20 km	assessment length
854,971	kg CO2e virgin
620,446	kg CO2e recycled
27.4%	lower modelled GWP

ROLE

Led project scope, openLCA modelling, input verification, scaling, sensitivity analysis, report integration, and team coordination.

MODEL

- Functional unit modelled per kilogram and scaled to the full road geometry.
- Density assumption: 2,200 kg/m3; transport distance: 50 km base case.
- ELCD 3.2 process proxies and ReCiPe 2016 impact assessment.
- Sensitivity checks covered transport, electricity mix, and recycled-concrete material loss.

EVIDENCE

The recycled-concrete scenario remained lower impact across global warming potential, cumulative energy demand, and acidification potential in the tested sensitivity cases.

DECISION SIGNAL

Under the stated assumptions, recycled concrete base provided the lower-impact option for the modelled road subbase.

LIMITATIONS

European proxy datasets were used where local datasets were unavailable. Results are comparative screening estimates, not a project environmental declaration.

Evidence basis: group report, openLCA model inputs, calculations, charts, sensitivity analysis, and presentation. Team work is identified clearly; the stated role reflects documented contributions.